Technical Reproducibility Validation Report

STRspy

Version v2.0
Autosomal STR (ONT)

Verification date	2026-11-01
Repository commit	dafdee7e7e5672c8dc732e8577dbe153f53a12f5
Result	**Not verified as documented**
Reached	Runs + Expected IO
Permalink	https://strhub.app/verified/strspy-ont
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

– Source available
– Installation successful
– End-to-end execution
– Output generated

NOT VERIFIED

– Genotype accuracy
– Concordance
– Forensic validity
– Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**Not verified as documented**
REACHED	Runs + Expected IO, 4/5 gates passed
WHY	The tool installed and its run produced its documented output.
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. What STRhub had to do to run this tool

STRhub wrote its own recipe for this tool, an environment and a command of its own rather than the repository's instructions, and ran that. Each item below is something a first-time user following the README would have to work out for themselves, and so a recommendation to the author. This run does not change the tool's label: the label is what happens as it is in the repository.

– Creates the conda environment straight from setup/STRspy_2.0_env.yml. Instead of: bash setup/STRspy_2.0.setup.sh, the documented installer. The installer stops at `read ans` for a confirmation; it hangs unattended.

– Unzips the prebuilt database db-v2/STRspy2.0-DB.zip and deletes the macOS files it ships with (__MACOSX, ._*, .DS_Store). Instead of: Nothing documented: the README does not mention the prebuilt database; the author offered it in issue #12. The stray files match the per-locus FASTA/BED globs and pollute the run.

– Writes config/ToolsConfig.txt directly, with the tool paths of the environment. Instead of: bash setup/GenerateToolConfig.sh followed by mv setup/UserToolsConfig.txt config/, as documented. The script writes UserToolsConfig.txt to the current directory, so the documented mv fails, and the wrapper expects ToolsConfig.txt.

– Runs src/STRspy_Normal_v2.0_Args.sh directly, with the flags its own -h lists. Instead of: bash ./STRspy_run_v2.0_Args.sh config/InputConfig.txt config/ToolsConfig.txt, the only documented command. The wrapper checks for the scripts in the repository root, where they are not (they are in src/), and exits: 'please make sure you are in the same directory of STRspy'.

With those changes, the run reached: **Runs + Expected IO**.

3. Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Runs	Executes end-to-end without crashing	PASS
Expected IO	Produces a non-empty file in the declared format	PASS
Plausible Output	Output structure matches expected genotype-bearing format	—
Execution log	strspy-ont.log-external.txt	

Stopping at a step is not a finding that the software is faulty. It records how far this particular attempt got. Software can stop early because a dependency it names is no longer available, because it expects input arranged differently from the reference sample, or because the automated environment cannot supply something it needs, such as commercially licensed components. The messages on screen when it stopped are reported below.

4. Errors Reported During the Run

The tool reported errors on 9 item(s) during the run. This does not assess whether the results produced are correct.

What happened	Times	Affected
Could not open expected files	18	D10S1248, D12S391, D13S317, D5S818, D7S820, D8S1179, FGA, TPOX, vWA

Structural errors, such as a file that will not open, an unrecognized command-line flag, or an incomplete build, do not depend on the sample: a coverage-limited slice yields fewer reads, but it cannot cause them. These are not attributable to STRhub's reference sample.

A small test file in the tool's own repository lets a new user run it on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's sample as well as STRhub's slice. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

5. What the Author Documents as a Known Issue

Quoted from the repository's README at the verified commit. STRhub did not establish any of this by running the tool; it is the author's own note about their own software.

Known bug (README line 291)

When using the wrapper ('STRspy_run_v1.0.sh'), One might run into an issue. The parallel version of **STRspy_Parallel_v2.0_Args.sh** is unable to properly connect with "gnu parallel" and exits the workflow without mapping or further analysis steps of the pipeline. Solution to this, the user can choose the Normal version to avoid any crash. ***Tip: Its good practice to use pre-aligned bams for quicker outcomes.***

6. Exact Run Command

This is the command that was executed, verbatim, in the CI environment.

```
# Environment: ubuntu-22.04 · STRspy v2.0 · commit dafdee7
$ bash /opt/strspy/src/STRspy_Normal_v2.0_Args.sh \
-s /data/in \
-r yes \
-t ont \
-f /opt/strspy/db/STRspy2.0-DB \
-b /opt/strspy/db/STRspy2.0-DB \
-l /opt/strspy/db-v2/STRspy_v2.DB.sort.bed \
-k 0.4 \
-o /data/out \
-c /opt/strspy/config/ToolsConfig.txt \
-d 1
```

7. Reproducibility Metadata

Tool	STRspy
Version	v2.0
Repository	https://github.com/unique379r/strspy
Commit (pinned)	dafdee7e7e5672c8dc732e8577dbe153f53a12f5
Container OS	ubuntu-22.04
Marker panel	Autosomal STR (ONT)
Verified on	2026-11-01 12:52:35 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/41307968157

Verified automatically, in a clean environment, on the tool's public source at the pinned commit. This certificate records what happened; it is not an endorsement by the tool's author.

8. Out of Scope

This report does not evaluate any of the following:

– Genotype correctness or accuracy
– Concordance against known truth sets
– Sensitivity, specificity, or stutter performance
– Allele calling accuracy or forensic casework suitability
– Regulatory compliance or ISO accreditation
– Multi-laboratory or multi-dataset reproducibility

9. Output Content Evidence

Output file	Countings/CSF1PO_input.bam_Allele_freqs.txt
Format	TSV
Sequence records	6
Distinct STR loci	0
Total reads	0
Max depth (single locus)	0

10. Verification Data

The repository ships example data (3 file(s)); this run did not use it. STRhub ran the verification using the public reference dataset listed below. A small test file in the repository lets a new user run the tool on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's own sample as well as this one. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

Dataset	1000 Genomes ONT, hg38 CODIS slice (R10 SUP)
Source	https://s3.amazonaws.com/1000g-ont/index.html?prefix=PROCESS…
License	Open access (1000 Genomes / HPRC). Research use. STRhub is not the data provider.
Ref. genome	GRCh38 / hg38
Loci tested	—

11. Verification Matrix

PASS	Reference dataset	1000 Genomes ONT, hg38 CODIS slice (R10 SUP)
	README check (5/5)	Advisory: all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

12. What This Run Needed Beyond the Repository

The result above describes a run configured as follows. Anyone repeating it needs the same things.

– Test data: no sample from the repository was used, so a public reference sample stood in.

– A container environment and a command written by STRhub, not taken from the repository's instructions; what they do differently is listed under "What STRhub had to do to run this tool".

13. Limitations

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

14. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

15. Conclusion

◆ **Runs end-to-end**

STRspy v2.0 installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit.

◆ **Produces output in the declared format**

The tool wrote a non-empty TSV file. Whether its contents look like genotype data was not established at this level.

◆ **Verdict: Runs**

The tool installed and its run produced its documented output.

◆ **Repository test data not used**

The repository ships example data (3 file(s)), which this run did not use: the gates above ran on STRhub's reference dataset. STRhub Verified supplied 1000 Genomes ONT, hg38 CODIS slice (R10 SUP) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

◆ **Reproducibility statement**

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.